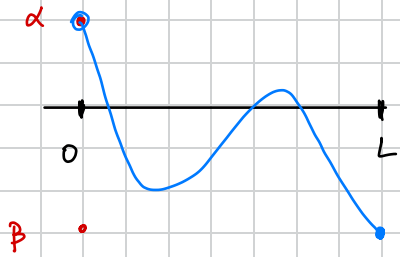
HOW TO IMPOSE DIRICHLET CONDITIONS

$$\begin{cases} \rho u - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial u}{\partial x}\right) = f & \text{in } (0,L) \\ u(0) = \alpha \qquad u(L) = \beta \end{cases} \qquad (P)$$



Find $u \in H^1(0,L)$ s.t.

$$\int_0^L \rho u v \, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} dx - \left[\mu \frac{\partial u}{\partial x} v\right]_0^L$$

$$-\mu(L)\frac{\partial u}{\partial x}(L)v(L) + \mu(0)\frac{\partial u}{\partial x}(0)v(0) = \int_0^L f v \, dx$$

$$\forall v \in H^1(0,L)$$

Choose $v = 0$ in $x=0$ & $x = L$

Find $\boxed{u \in H^1(0,L) \quad u(0) = \alpha \quad u(L) = \beta}$

$$\int_0^L \rho u v \, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} dx = \int_0^L f v \, dx \qquad \boxed{\forall v \in H^1(0,L) \quad v(0) = v(L) = 0}$$

$$H_0^1(0,L) = \{ v \in H^1(0,L) : v(0) = v(L) = 0 \}$$

If $\alpha = \beta = 0 \Rightarrow u, v \in H_0^1(0,L)$

How to recast (P) in such a way that the solution is 0 in $x=0$ & $x=L$?

- $w(x) = u(x) - R(x)$ where

  $R(x) \in H^1(0,L)$ s.t. $R(0) = \alpha$ & $R(L) = \beta$

  $\Rightarrow w(x) \in H^1(0,L)$ $\quad w(0) = w(L) = 0$

$\Rightarrow w(x) \in H_0^1(0,L)$

- $u(x) = w(x) + R(x)$

$$\begin{cases} \rho(w+R) - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial}{\partial x}(w+R)\right) = f & \text{in } (0,L) \\ w(0) = 0 \qquad w(L) = 0 \end{cases} \qquad (P')$$

$(P') \Longleftrightarrow (P)$

$$(\bullet)\ \rho w - \frac{\partial}{\partial x}\left(\mu \frac{\partial w}{\partial x}\right) = f - \rho R + \frac{\partial}{\partial x}\left(\mu \frac{\partial R}{\partial x}\right)$$

MULTIPLY (•) by $v \in H_0^1(0,L)$ and integrate

Find $w \in H_0^1(0,L)$ s.t.

$$\underbrace{\int_0^L \rho w v \, dx + \int_0^L \mu \frac{\partial w}{\partial x} \frac{\partial v}{\partial x} dx}_{a(w,v)} = \int_0^L f v \, dx \underbrace{- \int_0^L \rho R v \, dx - \int_0^L \mu \frac{\partial R}{\partial x} \frac{\partial v}{\partial x} dx}_{-a(R,v)} \quad \forall v \in H_0^1(0,L)$$

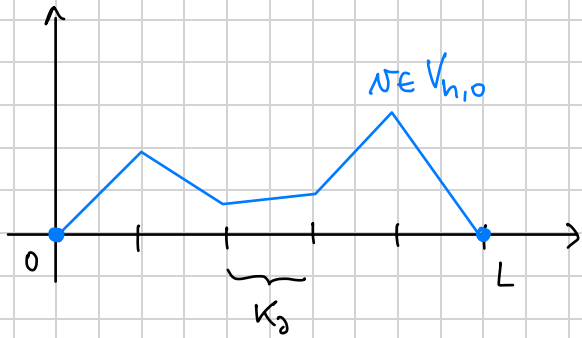
Find $w \in H_0^1(0,L)$ s.t.

$$a(w,v) = f(v) - a(R,v) \quad \forall v \in H_0^1(0,L)$$

FINITE ELEMENT FORMULATION

$V = H_0^1(0,L)$ $\qquad V_{h,0} \subset V$

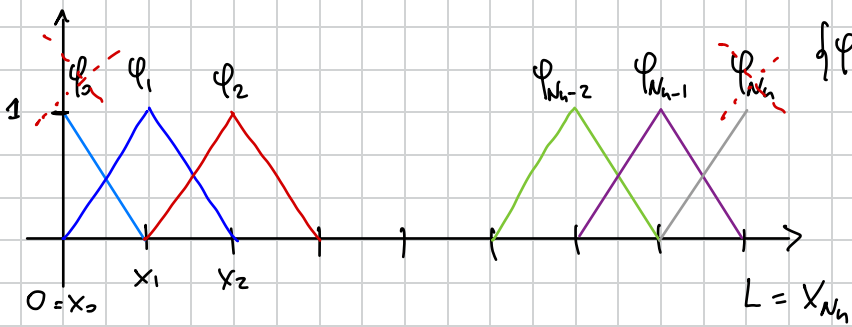
$$V_{h,0} = \left\{ v \in C^0(0,L) \text{ s.t. } v|_{K_j} \in \mathbb{P}^1(K_j) \ \& \ \underline{v(0) = v(L) = 0} \right\}$$



How to choose a basis for $V_{h,0}$?

$\{\varphi_j\}_{j=0}^{N_h}$ basis for V_h

$\{\varphi_j\}_{j=1}^{N_h-1}$ basis for $V_{h,0}$



$$w_h(x) = \sum_{j=1}^{N_{h-1}} w_j \varphi_j(x)$$

$$\dim(V_h) = N_h + 1 \quad \Rightarrow \quad \dim(V_{h,0}) = N_h - 1$$

$$\Rightarrow \quad A \longleftrightarrow a(\cdot,\cdot) \qquad A \in \mathbb{R}^{N_{h-1} \times N_{h-1}}$$

FE FORMULATION

Find $w_h \in V_{h,0}$ s.t.

$$a(w_h, v_h) = f(v_h) - a(R, v_h) \quad \forall v_h \in V_{h,0}$$

Find $\underline{w} \in \mathbb{R}^{N_h-1}$ s.t. $A\underline{w} = \underline{b} - A\underline{R}$

$\underline{b}, \underline{R} \in \mathbb{R}^{N_h-1}$

$\Rightarrow \underline{u} \in \mathbb{R}^{N_h+1}$? $\qquad \underline{u} = (\alpha, w_1, \dots, w_{N_h-1}, \beta)$

$R(x) \in V_h \quad R(0) = \alpha$

$R(L) = \beta$

$$\underline{R} = \begin{pmatrix} \alpha \\ 0 \\ \vdots \\ 0 \\ \beta \end{pmatrix} \in \mathbb{R}^{N_h+1}$$

$N_h = n \qquad u_1 = \alpha \quad u_2 = \beta$

$$\begin{cases} a_{11} u_1 + a_{12} u_2 + \dots + a_{1n} u_n = b_1 \\ a_{21} u_1 + a_{22} u_2 + \dots + a_{2n} u_n = b_2 \\ \vdots \\ a_{n-1,1} u_1 + a_{n-1,2} u_2 + \dots + a_{n-1,n} u_n = b_{n-1} \\ a_{n1} u_1 + a_{n2} u_2 + \dots + a_{nn} u_n = b_n \end{cases}$$

$$\underline{u} = (\alpha, u_1, \dots, u_{n-1}, \beta)$$

$$\begin{cases} u_1 = \alpha \\ a_{22} u_2 + \dots + a_{2n-1} u_{n-1} = b_2 - a_{21}\alpha - a_{2n}\beta \\ \\ a_{n-1,2} u_2 + \dots + a_{n-1\,n-1} u_{n-1} = b_{n-1} - a_{n-1,1}\alpha - a_{n-1\,n}\beta \\ u_n = \beta \end{cases}$$

$$A\underline{u} = \underline{b} - A\underline{R} + \underline{R}$$

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \vdots \\ \vdots & & A_{2:n-1,2:n-1} & \vdots \\ 0 & \cdots & \cdots & 0 \quad 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n-1} \\ u_n \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ b_2 \\ \vdots \\ b_{n-1} \\ 0 \end{pmatrix}}_{\underline{b}} - \underbrace{\begin{pmatrix} 0 \\ a_{21}\alpha + a_{2n}\beta \\ \vdots \\ a_{n-1,1}\alpha + a_{n-1\,n}\beta \\ 0 \end{pmatrix}}_{A\underline{R}} + \underbrace{\begin{pmatrix} \alpha \\ 0 \\ \vdots \\ 0 \\ \beta \end{pmatrix}}_{\underline{R}}$$

SOME RESULTS ABOUT CONVERGENCE ($\mathbb{P}^1$)

1) $$\| u - u_h \|_{H^1(0,L)}$$

$$= \left(\| u - u_h \|^2_{L^2(0,L)} + \left\| \frac{\partial u}{\partial x} - \frac{\partial u_h}{\partial x} \right\|^2_{L^2(0,L)} \right) \lesssim h$$

2) $$\| u - u_h \|_{L^2(0,L)} \lesssim h^2$$

3) $$\max_{x \in [0,L]} | u(x) - u_h(x) | \lesssim h^2$$

How do I verify in practice this behaviour?

VERIFICATION !

$\mathcal{T}_{h_1}$ is a mesh with size $h_1 \to E_1 = \| u - u_h^1 \|_{H^1(0,L)} \simeq h_1^p$

$\mathcal{T}_{h_2}$ is a mesh with size $h_2 \to E_2 = \| u - u_h^2 \|_{H^1(0,L)} \simeq h_2^p$

$$\frac{E_1}{E_2} \simeq \left(\frac{h_1}{h_2}\right)^p$$

$$p \approx \log\left(\frac{E_1}{E_2}\right) \Big/ \log\left(\frac{h_1}{h_2}\right)$$

I expect from theory that

$p \simeq 1$ if error $\|\cdot\|_{H^1(0,L)}$

$p \simeq 2$ if error $\|\cdot\|_{L^2(0,L)}$

WAVE EQ. WITH NEUMANN CONDITIONS

$$\begin{cases} \rho \dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial u}{\partial x}\right) = f & \text{in } (0,L)\times(0,T) \quad \circledast \\ \mu(0)\dfrac{\partial u}{\partial x}(0,t) = g_0(t) & t \in (0,T] \\ \mu(L)\dfrac{\partial u}{\partial x}(L,t) = g_L(t) & t \in (0,T] \\ u(x,0) = u_0(x) & x \in [0,L] \\ \dfrac{\partial u}{\partial t}(x,0) = u_1(x) & x \in [0,L] \end{cases}$$

Take $v(x) \in H^1(0,L)$, multiply $\circledast$, and integrate by parts

$$\int_0^L \rho \frac{\partial^2 u}{\partial t^2} v \, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} dx \overbrace{- \mu(L)\frac{\partial u}{\partial x}(L)}^{g_L(t)} v(L)$$

$$+ \underbrace{\mu(0)\frac{\partial u}{\partial x}(0)}_{g_0(t)} v(0) = \int_0^L f v \, dx$$

$\forall t \in (0,T]$ find $u(t) \in H^1(0,L)$ s.t.

$$\int_0^L \rho \frac{\partial^2 u}{\partial t^2} v \, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} dx = \int_0^L f v \, dx + g_L(t) v(L) - g_0(t) v(0) \quad \forall v \in H^1(0,L)$$

+ INITIAL CONDITIONS

WEAK FORM

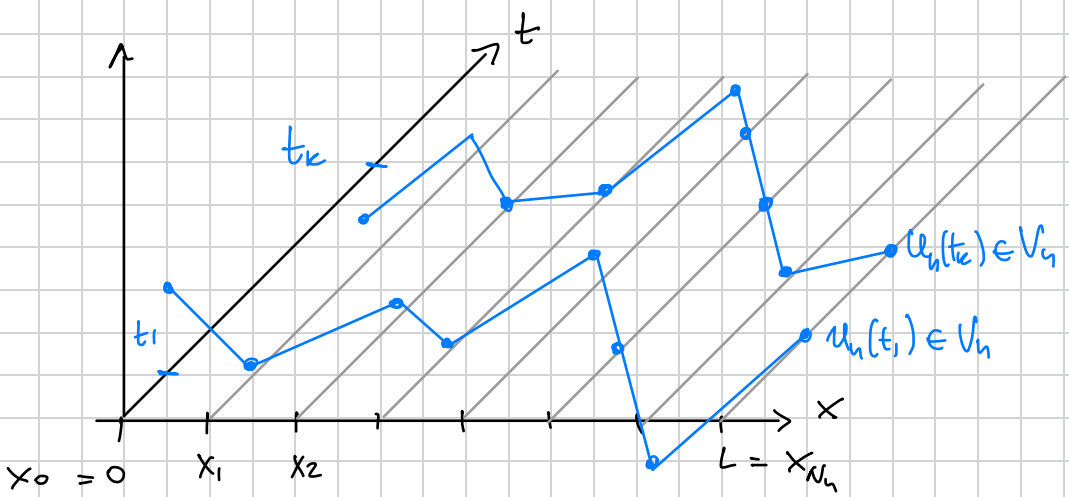
WELL-POSED (LAX-MILGRAM) $\Rightarrow$ $\exists!$ SOLUTION !

FINITE ELEMENT FORMULATION

$\forall t \in (0,T]$ $\quad u_h(t) \in V_h \subset H^1(0,L)$

mesh

$$V_h = \left\{ v \in C^0(0,L) : v|_{K_j} \in \mathbb{P}^1(K_j) \quad \forall K_j \in \mathcal{T}_h \right\}$$



$$u_h(x,t) = \sum_{j=0}^{N_h} \underbrace{u_j(t)}_{N_h+1 \text{ UNKNOWN COEFF.}} \varphi_j(x)$$

FE FORMULATION

$\forall t \in (0,T]$ find $u_h(t) \in V_h$ s.t.

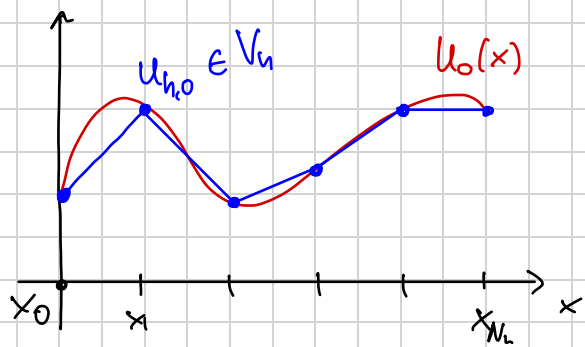
$$\underbrace{\int_0^L \rho \frac{\partial^2 u_h}{\partial t^2} v_h \, dx}_{m(\ddot{u}_h, v_h)} + \underbrace{\int_0^L \mu \frac{\partial u_h}{\partial x} \frac{\partial v_h}{\partial x} \, dx}_{a(u_h, v_h)} = \underbrace{\int_0^L f v_h \, dx + g_L(t) v(L) - g_0(t) v(0) \quad \forall v \in V_h}_{F(v_h)}$$

MASS

STIFFNESS

+ INITIAL CONDITIONS

$$\begin{cases} u_h(x,0) = u_{0,h}(x) \\ \dfrac{\partial u_h}{\partial t}(x,0) = u_{1,h}(x) \end{cases}$$



ALGEBRAIC FORMULATION

$$u_h(x,t) = \sum_{j=0}^{N_h} u_j(t)\,\varphi_j(x) \qquad v_h(x) = \varphi_i(x)$$

$$\ddot{u}_h(x,t) = \sum_{j=0}^{N_h} \ddot{u}_j(t)\,\varphi_j(x)$$

$$m\left(\sum_{j=0}^{N_h} \ddot{u}_j(t)\,\varphi_j(x), \varphi_i(x)\right) \qquad \forall i = 0,\dots,N_h$$

$$+\, a\left(\sum_{j=0}^{N_h} u_j(t)\,\varphi_j(x), \varphi_i(x)\right) = F(\varphi_i)$$

$$\sum_{j=0}^{N_h}\Big[\underbrace{m(\varphi_j,\varphi_i)}_{m_{ij}}\,\ddot{u}_j(t) + \underbrace{a(\varphi_j,\varphi_i)}_{a_{ij}}\,u_j(t)\Big] = \underbrace{F(\varphi_i)}_{b_i}$$

$$m_{ij} = \int_0^L \rho\,\varphi_j\,\varphi_i\,dx \;\longleftrightarrow\; M$$

$$a_{ij} = \int_0^L \mu\,\frac{\partial\varphi_j}{\partial x}\frac{\partial\varphi_i}{\partial x}\,dx \;\longleftrightarrow\; A$$

$$b_i = \int_0^L f\,\varphi_i\,dx + g_L(t)\,\varphi_i(L) - g_0(t)\,\varphi_i(0)$$

$$\begin{cases} M\,\ddot{\underline{u}}(t) + A\,\underline{u}(t) = \underline{b}(t) & \forall t \in (0,T] \\ \underline{u}(0) = \hat{\underline{u}}_0 \\ \dot{\underline{u}}(0) = \hat{\underline{u}}_1 \end{cases}$$

SYSTEM OF LINEAR ORDINARY DIFF. EQs

$$\Delta T = \frac{T}{M} \qquad t_i = t_0 + i\,\Delta T$$

$$M\,\underbrace{\ddot{\underline{U}}(t_k)} + A\,\underline{U}(t_k) = \underline{b}(t_k)$$

$$\ddot{U}(t_k) \simeq \underbrace{\frac{\underline{U}(t_{k+1}) - 2\underline{U}(t_k) + \underline{U}(t_{k-1})}{\Delta T^2}}_{\text{TAYLOR EXPANSION}}$$

2ND ORDER APPROX $\simeq \Delta T^2$

$$\overbrace{U(t_k + \Delta T)}^{U(t_{k+1})} = U(t_k) + \Delta T\,\dot{U}(t_k) + \frac{\Delta T^2}{2}\ddot{U}(t_k) + \dots$$

$$\underbrace{U(t_k - \Delta T)}_{U(t_{k-1})} = U(t_k) - \Delta T\,\dot{U}(t_k) + \frac{\Delta T^2}{2}\ddot{U}(t_k)$$

SUM 2 EQs :

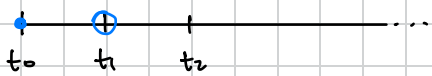
$$\Delta T^2\,\ddot{U}(t_k) = U(t_{k+1}) - 2U(t_k) + U(t_{k-1})$$

$$\underline{U}_k = \underline{U}(t_k)$$

$$M\left(\frac{U_{k+1} - 2U_k + U_{k-1}}{\Delta T^2}\right) + A\,\underline{U}_k = \underline{b}_k$$

$$\boxed{M\,U_{k+1} = \Delta T^2\,\underline{b}_k - \left(\Delta T^2 A - 2M\right)\underline{U}_k - M\,\underline{U}_{k-1}}$$

GENERAL STEP (B)



→ MODIFY THE DISCRETIZATION

$$\underbrace{U(t_1)}_{\underline{U}_1} = \underbrace{U(t_0)}_{\hat{\underline{U}}_0} + \Delta T\,\underbrace{\dot{U}(t_0)}_{\hat{\underline{U}}_1} + \frac{\Delta T^2}{2}\ddot{U}(t_0)$$

INITIAL CONDITIONS

$$\ddot{\underline{U}}(t_0) \simeq \frac{2}{\Delta T^2}\left(\underline{U}_1 - \hat{\underline{U}}_0 - \hat{\underline{U}}_1\,\Delta T\right)$$

$$\frac{2}{\Delta T^2}\,M\left(\underline{U}_1 - \hat{\underline{U}}_0 - \Delta T\,\hat{\underline{U}}_1\right) = \underline{b}_0 - A\,\hat{\underline{U}}_0$$

$$\boxed{M\,\underline{U}_1 = \frac{\Delta T^2}{2}\underline{b}_0 - \left(\frac{\Delta T^2}{2}A - M\right)\hat{\underline{U}}_0 + \frac{\Delta T}{2}M\,\hat{\underline{U}}_1}$$

INITIAL STEP (A)

(A) + (B) $\Rightarrow$ LEAP-FROG SCHEME

- 2ND ACCURATE IN TIME $E \sim \Delta T^2$
- EXPLICIT SCHEME
  (ONLY M HAS TO BE INVERTED)
- STABILITY CONDITION (CFL)

$$\Delta T \leq c \frac{h}{c}$$

h — MESH SIZE

c — WAVE VEL $\quad c = \sqrt{\frac{\mu}{\rho}}$

$c \in (0,1)$